

Structure-Based Prediction of Autophagy Modulators: From Transcriptome Collapse to Morgan Fingerprint Classifiers

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Abstract

Autophagy is a conserved cellular quality-control pathway whose pharmacological modulation is therapeutically attractive but hard to screen computationally: the observable signal sits several steps downstream of small-molecule binding, and perturbational transcriptomes confound pathway-specific modulation with generic stress responses. We present a scaffold-aware empirical audit of four formulations of the task (full transcriptome regression, gene-signature matching, binary modulator classification, and activator-versus-inhibitor direction prediction) using LINCS L1000 transcriptomics, the Human Autophagy Modulator Database (HAMdb), and a 99,433-compound ChEMBL inference library. A Uni-Mol neural network trained to predict 12,328-gene expression responses collapsed on canonical opposing controls (Rapamycin versus Wortmannin, Spearman $\rho = +0.97$ versus $+0.43$ observed), and genome-wide diagnostics found no detectable structure-expression signal. Reframed as binary modulator classification, XGBoost on 2,048-bit Morgan fingerprints achieved compound-level AUROC 0.845 with L1000 context and 0.84–0.88 (95% bootstrap CI) without context, ranking held-out controls correctly. Scoring the ChEMBL library enriched autophagy and mTOR/PI3K annotations up to $\sim 36\times$ and $\sim 58\times$ baseline at stringent thresholds. Direction prediction remained near random (macro F1 0.532), limited by 132 curated inhibitors. Binary structure-based screening is reliable and externally valid; transcriptome regression and structure-only direction prediction are not.

Keywords: autophagy; machine learning; LINCS L1000; Morgan fingerprints; XGBoost; virtual screening; perturbation modelling

Introduction

Autophagy is a conserved lysosomal degradation and recycling program with fundamental roles in aging, neurodegeneration, cancer, and metabolic disease, and both its activation and its inhibition are of therapeutic interest [1, 2]. Pharmacological modulation of autophagy is non-trivial: approved and experimental modulators act through heterogeneous and often indirect mechanisms (mTOR and PI3K/Vps34 pathway inhibition, lysosomotropic weak bases that block autophagic flux, and direct regulators of the autophagy machinery), and no single assay class cleanly separates induction from blockade [3, 4]. Reliable computational prioritization of autophagy modulators would therefore be valuable, but it is unclear which modelling formulation can deliver it.

A dominant recent strategy has been to learn end-to-end mappings from chemical structure to genome-wide transcriptional responses, trained on large perturbational compendia such as the LINCS L1000 platform [6]. Successive models (DLEPS, DeepCE, CIGER, TranSiGen, and MiTCP) have reported improving accuracy on this structure-to-expression task [8, 9, 10, 11, 12], and related architectures predict responses to genetic perturbations at single-cell resolution [13, 14]. Yet the reliability of these models has been questioned: a recent community benchmark concluded that deep-learning perturbation models do not yet outperform

simple linear baselines on gene-expression prediction [15], consistent with the broader pattern that learned molecular representations often fail to beat well-tuned fingerprints on ligand-based tasks [16].

On the ligand-based side, molecular representations and evaluation methodology are mature. Extended-connectivity (Morgan) fingerprints remain strong, cheap baselines for bioactivity prediction [17]; pretrained 3D-aware (Uni-Mol) and language-aligned (CLAMP) embeddings offer learned alternatives [18, 19]; and it is well established that random splits inflate apparent performance through scaffold leakage, motivating Murcko-scaffold splits and memorization-sensitive evaluation [21, 22, 23]. For imbalanced classification, precision-recall behaviour and per-class summaries are preferred over raw accuracy [24], and virtual-screening utility is best expressed as early-recognition enrichment over a defined baseline [25].

Where does autophagy fit in this landscape? Machine learning has been applied to autophagy mostly through imaging-based detection and biomarker prediction, and a recent review catalogues the field’s open questions rather than mature solutions [26]. The closest prior prioritization effort used HAMdb compounds within a heterogeneous interaction network via random-walk propagation, not molecular structure [27]. To our knowledge, no prior work has trained a ligand-based classifier of autophagy modulators from chemical structure, and no prior work has audited whether structure-to-transcriptome models can capture autophagy pharmacology at all. Here we provide both, under a single scaffold-aware evaluation protocol, and arrive at a clear practical recommendation.

Results

A structure-to-transcriptome model collapses on opposing controls

We trained a neural network to predict full 12,328-gene L1000 log-fold-change vectors from 512-dimensional Uni-Mol 3D molecular embeddings plus cell-line identity (“V6” model; ~18,000 Level-5 signatures). As a functional audit we scored two canonical controls with opposing autophagy biology: Rapamycin (allosteric mTOR inhibitor; induces autophagy) and Wortmannin (PI3K inhibitor; blocks autophagy induction). The model predicted near-identical profiles for the two compounds (per-gene Spearman $\rho = +0.970$; Figure 1a), with predictions collapsed onto the identity line and, additionally, a ~10-fold compression of dynamic range (prediction range $\sim \pm 0.2$ logFC versus $\sim \pm 3$ observed). This is the signature of an invariant mean-expression output, not pathway-specific response prediction. Ground-truth profiles of the same two compounds at the same dose/time/cell context correlate only $\rho = +0.434$ (Figure 1b); opposing pharmacologies are partially separable in the transcriptome, but the model inflates their similarity and flattens their differences.

Three diagnostics indicate the failure is intrinsic to the task formulation rather than to a specific architecture. First, genome-wide association between molecular structure and observed expression was undetectable: Pearson correlation between Uni-Mol embeddings and expression profiles was -0.0097 , and zero of 12,328 genes reached $|\text{Spearman}| > 0.2$ against structure. Second, ridge-regression upper bounds were negative ($R^2 \approx -0.11$ to -0.16), so no linear structure-expression mapping exists on a generic compound panel. Third, a training-set audit showed the 2,000-compound smoke-test panel contained essentially no annotated autophagy pharmacology (one mTOR inhibitor; zero PI3K inhibitors); restricting training to the 227 HAMdb modulators with L1000 signatures improved mean per-gene Spearman by 61% while using 90× fewer compounds; whatever signal exists is concentrated in annotated modulators, not the general transcriptome.

Binary modulator classification is robust and externally valid

Reframing the objective as binary classification (autophagy modulator versus non-modulator) restored robust performance. For *context-aware* models, positives were 227 HAMdb modulators with matching L1000 signatures and negatives were matched random L1000 signatures (9,804 train / 3,856 test signatures from 137 held-out compounds; 70/30 Murcko-scaffold-aware split; mTOR/PI3K-overlapping compounds excluded;

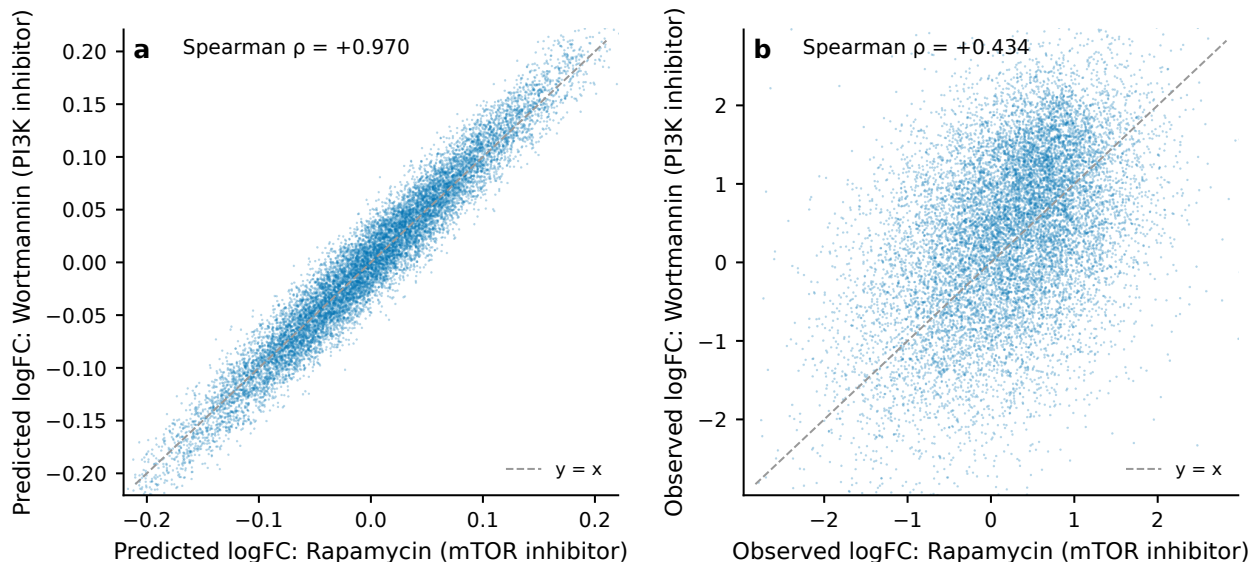


Figure 1. Transcriptome-regression collapse on opposing biological controls. (a) Predicted log-fold-change of Rapamycin versus Wortmannin across all 12,328 genes (one point per gene; Spearman $\rho = +0.970$). (b) Ground-truth L1000 profiles of the same two compounds (A549, 6 h, highest dose; Spearman $\rho = +0.434$). The model inflates the similarity between opposing pharmacologies and compresses the response dynamic range roughly tenfold.

Rapamycin and Chloroquine held out). Morgan fingerprints consistently outperformed CLAMP embeddings here, and XGBoost outperformed its linear ridge counterpart (Table 1). The best context-aware model (XGBoost, Morgan + context) reached compound-level AUROC 0.845 (compound-level probabilities are the mean of per-signature probabilities within a compound) and AUPRC 0.810, and ranked the held-out canonical controls at the top of the score distribution (Rapamycin 0.981; Chloroquine 0.992).

Table 1. Context-aware binary modulator classification (L1000 signatures + cell/dose/time context; scaffold-aware held-out test of 137 compounds). Sig., signature level.

Model	Features	Sig. AUROC	Compound AUROC	AUPRC	Rapamycin	Chloroquine
Ridge	Morgan + context	0.705	0.823	0.765	—	—
Ridge	CLAMP + context	0.560	0.722	0.654	—	—
XGBoost	CLAMP + context	0.669	0.800	0.749	0.706	0.548
XGBoost	Morgan + context	0.765	0.845	0.810	0.981	0.992

Removing L1000 context entirely and training on chemical structure alone (646 HAMdb modulators versus 5,893 scaffold-family-balanced ChEMBL neutrals; 20/80 scaffold-aware split) preserved strong discrimination: XGBoost on Morgan fingerprints reached AUROC 0.858 (AUPRC 0.642) on the original training split, and adding CLAMP embeddings improved this slightly (AUROC 0.866; AUPRC 0.629), the reverse of the context-aware comparison, suggesting that CLAMP contributes bioassay-related chemical signal that is redundant once cellular context is available.

Because single scaffold splits carry substantial sampling variance, we rebuilt the split deterministically (PYTHONHASHSEED-fixed scaffold clustering, seed 42), retrained both classifiers on the reconstructed training portion, and evaluated them on 1,365 held-out compounds with 1,000-fold bootstrap confidence intervals (the evaluation underlying Figure 2): Morgan-only AUROC 0.877 (95% CI [0.843, 0.909]) and Morgan + CLAMP 0.839 (95% CI [0.799, 0.875]). The intervals overlap, so on this test scale the two models are statistically indistinguishable, in contrast to the original split, a reminder that per-split AUROC differences

of a few points should not be over-interpreted. Consistent with this, evaluating the original saved models (trained on a different scaffold split) on this test set inflated AUROC to 0.97: cross-split scaffold overlap is partially memorized, which is precisely the leakage failure mode that scaffold splitting is designed to prevent and why models and splits must be reported jointly, with intervals.

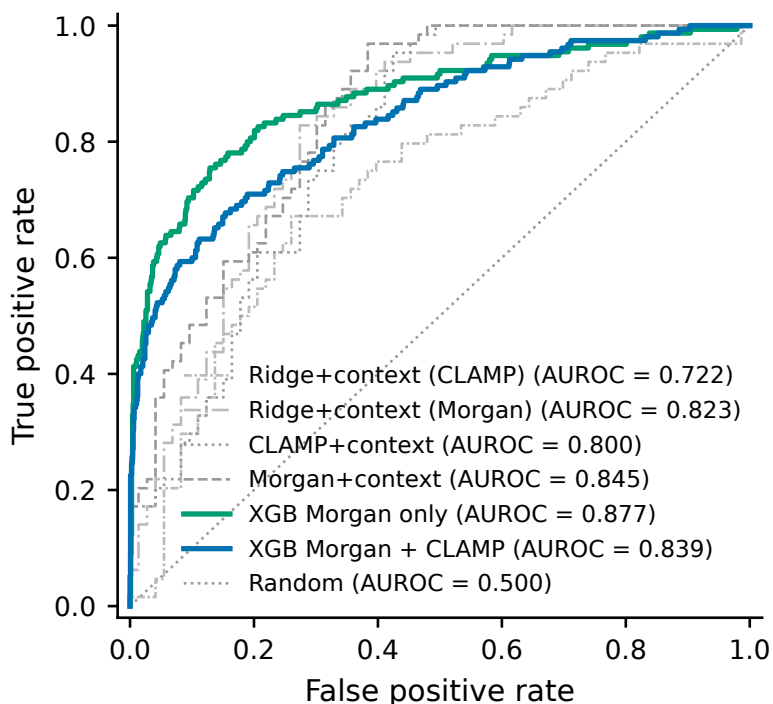


Figure 2. ROC curves for binary modulator classification under scaffold-aware validation. Morgan fingerprints lead the context-aware models, and the structure-only XGBoost classifiers perform best overall (reconstructed evaluation split; AUROC with 95% bootstrap intervals in text). Ridge baselines and the random classifier are shown for reference.

External validation on 99,433 ChEMBL compounds

We scored 99,433 neutral ChEMBL small molecules (deduplicated from 99,434 candidates; none in training) with both structure-only classifiers and measured enrichment of orthogonal bioactivity annotations: autophagy-assay activity and mTOR/PI3K-pathway target engagement (ChEMBL assay/target records). Library-wide baselines are 0.18% and 0.24%, respectively. Enrichment rises monotonically with score threshold for both models, with 95% binomial confidence intervals that exclude baseline from 0.70 upward for Morgan + CLAMP (Figure 3). At the most stringent threshold (≥ 0.95 ; Table 2), Morgan + CLAMP recovers 6.41% autophagy-active and 13.99% mTOR/PI3K-annotated compounds among only 343 prioritized structures ($\sim 36\times$ and $\sim 58\times$ baseline), outperforming Morgan-only (1.61% and 3.23% among 310; $\sim 9\times$ and $\sim 13\times$) by $\sim 4\times$ on both axes. At this threshold the 95% binomial (Wilson) intervals of the two models are disjoint for both annotation types (autophagy: [4.3, 9.5]% versus [0.7, 3.7]%; mTOR/PI3K: [10.7, 18.1]% versus [1.8, 5.8]%), so the CLAMP-driven improvement is statistically credible despite modest absolute counts. The top-200 hits of the two models overlap by only 36 compounds: CLAMP shifts prioritization toward a distinct, more biologically annotated chemical space. These enrichments are conservative lower bounds, because many true modulators carry no ChEMBL annotation.

Table 2. ChEMBL annotation enrichment at modulator-score thresholds. Baselines: 0.18% autophagy-active, 0.24% mTOR/PI3K across the full library. EF, enrichment factor over the relevant baseline.

Threshold	<i>n</i>	Model	% lib.	Autophagy %	mTOR/PI3K %	EF (auto.)	EF (mTOR)
≥ 0.50	2,791	Morgan+CLAMP	2.8	1.61	3.08	8.9×	12.8×
≥ 0.50	21,897	Morgan only	22.0	0.40	0.51	2.2×	2.1×
≥ 0.80	998	Morgan+CLAMP	1.0	3.31	6.81	18.4×	28.4×
≥ 0.80	4,314	Morgan only	4.3	0.72	1.07	4.0×	4.5×
≥ 0.95	343	Morgan+CLAMP	0.34	6.41	13.99	35.6×	58.3×
≥ 0.95	310	Morgan only	0.31	1.61	3.23	8.9×	13.5×

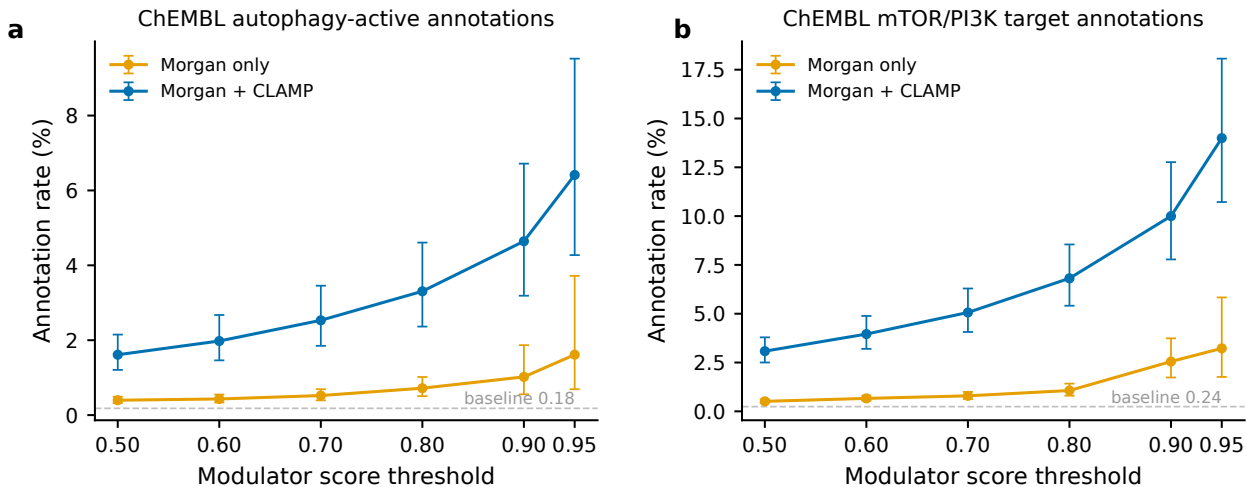


Figure 3. External validation on 99,433 ChEMBL compounds. Annotation rate (a: autophagy-active; b: mTOR/PI3K pathway targets) versus modulator-score threshold, with 95% binomial confidence intervals and library-wide baselines (dashed). Morgan + CLAMP dominates at every threshold, reaching 6.41% and 13.99% at ≥ 0.95 .

Activator-versus-inhibitor direction prediction fails

Binary screening succeeds; directional classification does not. A flat 3-class model (activator / inhibitor / neutral) achieved deceptively high accuracy (0.900) but macro F1 of only 0.532 (Morgan-only) and 0.455 (Morgan + CLAMP); per-class inspection shows the score is driven almost entirely by the dominant neutral class, with inhibitor discrimination particularly poor (Figure 4). A cascaded two-stage design (binary screen, then a dedicated direction model) confirmed the limitation: 5-fold scaffold-aware cross-validation of the direction stage gave AUROC 0.534 ± 0.045 , AUPRC 0.366 ± 0.058 , F1 0.326 ± 0.048 (per-fold AUROCs 0.478–0.598), with held-out test AUROC 0.498, at or below random. No representation rescued the task: class-balanced Morgan sampling was best (AUROC 0.594 ± 0.060) and physicochemical descriptors were worse (0.454). The two independent direction models agreed on only 55.9% of ChEMBL predictions.

The failure is best understood as a data problem compounded by intrinsic ambiguity. After scaffold filtering only 132 curated inhibitors remain, an insufficient basis for generalizable structure–direction rules, and the two classes genuinely overlap in chemical space: lysosomotropic weak-base inhibitors (e.g. Chloroquine) and kinase-targeting activators (e.g. mTOR inhibitors) are not cleanly separable by substructure. Direction labels should therefore not drive prioritization until substantially larger, scaffold-diverse inhibitor sets exist.

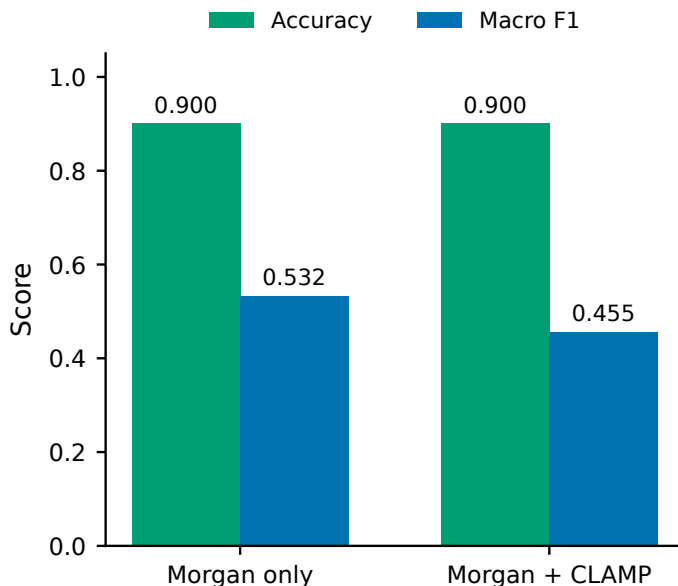


Figure 4. Direction prediction is dominated by class imbalance. Accuracy and macro F1 for flat 3-class classifiers. High accuracy reflects the majority neutral class; macro F1 (0.532 Morgan-only, 0.455 Morgan + CLAMP) shows activators and inhibitors are not reliably separable from structure alone.

Gene-signature matching captures generic stress, not autophagy flux

As an intermediate formulation we constructed consensus autophagy signatures (a 199-gene pathway set and a refined 25-gene flux-contrastive core; see Methods) and ranked compounds by signature correlation. The same confounder appeared at pathway level. An activator consensus was highly split-half reproducible ($\rho = 0.971 \pm 0.006$), yet individual compound signatures correlated with it only 0.071 ± 0.270 , while *random* compounds correlated 0.713 ± 0.032 : the signature largely encodes generic stress. The inhibitor and activator consensus signatures correlated 0.63 with each other, consistent with compensatory autophagy transcription under lysosomal blockade, and naive inverse search enriched random chemistry ($\sim 15\times$ at top-50). A rigorous redesign (conservative curation, per-cell-line contrastive signatures, dose windowing, permutation nulls, false-discovery control) *reduced* top-200 enrichment to 2–2.5%, and only 5 of the top 200 flux-signature hits carried ChEMBL autophagy-assay annotations. We do not recommend signature matching for autophagy prioritization on L1000 data.

What the structure-only classifier learned

Per-bit XGBoost gains from the saved Morgan-only model (1,155 of 2,048 bits carry nonzero gain) decode to chemically plausible discriminative environments (Figure 5): aliphatic amine and ether motifs (3.3-fold enriched in modulators), diazine/purine-like heteroaromatics (2.9-fold), thioether-linked aromatics, and an ester/acid motif (6.2-fold). The single most enriched feature is a bare chlorine environment (10.3-fold). High gain and high enrichment do not coincide everywhere (a trifluoromethyl-adjacent environment ranks high by gain yet is *depleted* in modulators), showing the model exploits absence as well as presence patterns. Because gain-based importance is biased toward high-cardinality features and Morgan bits are hashed, these decoded SMARTS are representative atom environments, not unique substructure assignments.

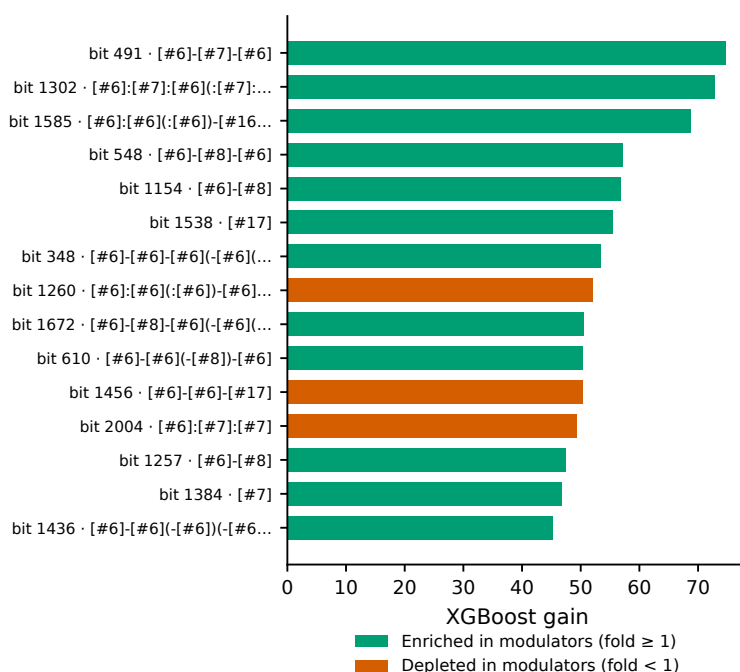


Figure 5. Top discriminative Morgan fingerprint bits of the structure-only classifier. Bars ranked by XGBoost gain; colour encodes fold-enrichment of bit occurrence in modulators versus a 2,000-compound ChEMBL background (green: enriched; orange: depleted).

Discussion

This audit’s negative results are as informative as its positive ones, and each connects to a known failure mode in the literature. The V6 collapse is a concrete instance of what community benchmarks have begun to document for perturbation models [15]: high-capacity models can exhibit smooth training dynamics while learning an invariant mean response. Our diagnostics localize the cause upstream of architecture (no marginal structure–expression association is detectable on a generic compound panel) and downstream of validation methodology: the collapse was invisible to random-split metrics and was exposed only by held-out biological controls and by correlating the *predictions* of opposing compounds. We recommend this pairwise-control audit as standard practice for any model claiming pathway-specific transcriptome prediction. It is also worth noting that ground-truth opposing controls still share appreciable transcription ($\rho = +0.43$ at 6 h); generic stress co-occurs with real signal, which is precisely why mean-output collapse is so easy to mistake for success.

The success of Morgan-fingerprint classifiers is consistent with molecular-representation benchmarks in which simple fingerprints remain hard to beat [16, 17]. CLAMP’s added value is real but context-dependent: indistinguishable from Morgan-only on the 1,365-compound held-out test, yet clearly superior at external scale on 99,433 ChEMBL compounds (Figure 3); small held-out sets simply lack the power to resolve differences that matter at screening scale. Two caveats qualify the external validation. First, CLAMP was pretrained on ChEMBL bioassay text–molecule pairs, so part of its ChEMBL enrichment could reflect pre-training exposure rather than pure generalization; the Morgan-only model, which shows no such overlap and still enriches ~9–13 $\times$, provides an unaffected control. Second, enrichment is measured against incomplete annotations; the true enrichment is higher but unquantifiable without prospective experimental testing.

For autophagy practice, the binary modulator score is the robust output of this pipeline. The failure of direction prediction is a data problem before it is a modelling problem: 132 inhibitors cannot define a

boundary in fingerprint space, and lysosomotropic inhibitors and kinase-targeting activators overlap chemically. Likewise, L1000 mRNA cannot separate autophagy induction from lysosomal blockage [4], and our flux-focused signature analysis showed conserved stress responses dominate the pathway-level signal. Future work should therefore train on direct flux readouts (LC3-II turnover, SQSTM1/p62 degradation, lysosomal flux imaging) as labels, expand scaffold-diverse inhibitor sets, and prospectively test the high-confidence ChEMBL prioritizations this pipeline produces.

Methods

Data resources

HAMdb [5] provided 841 curated autophagy modulators; after filtering to compounds with valid canonical SMILES and assignable direction, 514 activators and 132 inhibitors (plus 167 neutral) were retained (rule-based curation of the Category/pathway fields; `hamdb_autophagy_directions.csv`).

LINCS L1000 Phase I (GSE92742; Level-5 COMPZ.MODZ, $n473647 \times 12328$) [6] yielded 471,660 valid signatures across 20,315 compounds, 5,509 unique Murcko scaffolds, and 76 cell lines (6–96 h exposures). Of 813 curated HAMdb compounds, 222 (~26%) matched L1000 by PubChem CID; 227 modulators had usable signatures for the context-aware task.

Autophagy gene sets: a 199-gene pathway set (KEGG hsa04140 + literature) and a 25-gene flux-contrastive core (`curate_autophagy_genes.py`).

ChEMBL [7]: (i) a neutral inference library of 99,434 candidate small molecules (99,433 scored after SMILES de-duplication), pre-filtered to exclude autophagy/mTOR/PI3K/lysosomal annotations; and (ii) assay/target annotations for enrichment analysis. A scaffold-family-balanced subset of 5,893 neutrals served as training negatives.

Molecular representations

2,048-bit Morgan circular fingerprints (RDKit 2026.03, radius 2) [17, 28]; 768-dimensional CLAMP embeddings from the pretrained checkpoint [19]; 512-dimensional Uni-Mol 3D-aware embeddings [18]. Context features for L1000-context models: one-hot cell line, $\log_{10}$ dose, one-hot exposure time.

Models and training

The V6 transcriptome model was a neural network mapping Uni-Mol embeddings to 12,328-gene log-fold-change vectors. Classifiers used XGBoost (binary or multi softmax objective with `scale_pos_weight` class balancing; up to 1,000 trees, depth 6) [20], ridge logistic regression (class-weighted), and multilayer perceptrons. The cascade chained the binary Morgan-only classifier (stage 1) with a dedicated Morgan-only direction classifier (stage 2). All random seeds were fixed (42).

Validation and statistics

All splits were Murcko-scaffold-aware [21, 22] (context models 70/30; structure-only 20/80) to avoid scaffold leakage [23]. Canonical controls (Rapamycin, Wortmannin, Chloroquine) were held out and used as biological audits. Imbalanced classification was summarized with macro F1 and precision–recall metrics rather than accuracy alone [24]. Uncertainty: AUROC point estimates on the reconstructed structure-only split are accompanied by 1,000-fold bootstrap 95% confidence intervals (resampling test compounds); annotation rates carry 95% binomial (Wilson) intervals. Enrichment factors follow early-recognition convention [25]: observed annotated fraction at threshold divided by the library-wide baseline fraction.

Feature-importance analysis

For the saved structure-only Morgan model, per-bit XGBoost gains were decoded to atom-environment SMARTS via RDKit bit information across the 646 HAMdb modulators; fold-enrichment of bit occurrence was estimated against a 2,000-compound random ChEMBL background sample (seed 42).

Software environment

Python 3.12; RDKit 2026.03; XGBoost 3.4.1; PyTorch 2.14 (CPU); CLAMP via its official checkpoint; matplotlib for figures.

Data and code availability

HAMdb: <http://hamdb.scbdd.com>. LINCS L1000 Phase I: GEO GSE92742. ChEMBL: <https://www.ebi.ac.uk/chembl/>. The complete benchmarking pipeline, trained model weights, result tables (binary_models_comparison.csv, chembl_enrichment_morgan_vs_clamp.csv, paper_feature_importance.csv, bootstrap_auroc_ci.csv), and figure-generation code are available at <https://github.com/Abolfazl-Alipour/NAFS> (branch master) and archived at <https://doi.org/10.5281/zenodo.22986834>.

Competing interests

The author declares no competing interests.

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References

- [1] Levine, B. & Kroemer, G. Biological functions of autophagy genes: a disease perspective. *Cell* **176**, 11–42 (2019). doi:10.1016/j.cell.2018.09.048
- [2] Klionsky, D. J. *et al.* Autophagy in major human diseases. *EMBO J.* **40**, e108863 (2021). doi:10.15252/embj.2021108863
- [3] Galluzzi, L., Bravo-San Pedro, J. M., Levine, B., Green, D. R. & Kroemer, G. Pharmacological modulation of autophagy: therapeutic potential and persisting obstacles. *Nat. Rev. Drug Discov.* **16**, 487–511 (2017). doi:10.1038/nrd.2017.22
- [4] Klionsky, D. J. *et al.* Guidelines for the use and interpretation of assays for monitoring autophagy (4th edition). *Autophagy* **17**, 1–382 (2021). doi:10.1080/15548627.2020.1797280
- [5] Wang, N.-N. *et al.* HAMdb: a database of human autophagy modulators with specific pathway and disease information. *J. Cheminform.* **10**, 34 (2018). doi:10.1186/s13321-018-0289-4
- [6] Subramanian, A. *et al.* A Next Generation Connectivity Map: L1000 platform and the first 1,000,000 profiles. *Cell* **171**, 1437–1452 (2017). doi:10.1016/j.cell.2017.10.049

- [7] Mendez, D. *et al.* ChEMBL: towards direct deposition of bioassay data. *Nucleic Acids Res.* **47**, D930–D940 (2019). doi:10.1093/nar/gky1075
- [8] Zhu, J. *et al.* Prediction of drug efficacy from transcriptional profiles with deep learning (DLEPS). *Nat. Biotechnol.* **39**, 1444–1452 (2021). doi:10.1038/s41587-021-00946-z
- [9] Pham, T.-H., Qiu, Y., Zeng, J., Xie, L. & Zhang, P. A deep learning framework for high-throughput mechanism-driven phenotype compound screening and its application to COVID-19 drug repurposing (DeepCE). *Nat. Mach. Intell.* **3**, 247–257 (2021). doi:10.1038/s42256-020-00285-9
- [10] Pham, T.-H. *et al.* Chemical-induced gene expression ranking and its application to pancreatic cancer drug repurposing (CIGER). *Patterns* **3**, 100441 (2022). doi:10.1016/j.patter.2022.100441
- [11] Tong, X. *et al.* Deep representation learning of chemical-induced transcriptional profile for phenotype-based drug discovery (TranSiGen). *Nat. Commun.* **15**, 5378 (2024). doi:10.1038/s41467-024-49620-3
- [12] Yang, K., Cheng, J., Cao, S., Pan, X., Shen, H.-B. & Yuan, Y. Predicting transcriptional changes induced by molecules with MiTCP. *Brief. Bioinform.* **26**, bbaf006 (2025). doi:10.1093/bib/bbaf006
- [13] Lotfollahi, M., Wolf, F. A. & Theis, F. J. scGen predicts single-cell perturbation responses. *Nat. Methods* **16**, 715–721 (2019). doi:10.1038/s41592-019-0494-8
- [14] Roohani, Y., Huang, K. & Leskovec, J. Predicting transcriptional outcomes of novel multigene perturbations with GEARS. *Nat. Biotechnol.* **42**, 927–935 (2024). doi:10.1038/s41587-023-01905-6
- [15] Ahlmann-Eltze, C., Huber, W. & Anders, S. Deep-learning-based gene perturbation effect prediction does not yet outperform simple linear baselines. *Nat. Methods* **22**, 1657–1661 (2025). doi:10.1038/s41592-025-02772-6
- [16] Praski, M., Adamczyk, J. & Czech, W. Benchmarking pretrained molecular embedding models for molecular representation learning. *arXiv:2508.06199* (2025).
- [17] Rogers, D. & Hahn, M. Extended-connectivity fingerprints. *J. Chem. Inf. Model.* **50**, 742–754 (2010). doi:10.1021/ci100050t
- [18] Zhou, G., Gao, Z., Ding, Q., Zheng, H., Xu, H., Wei, Z., Zhang, L. & Ke, G. Uni-Mol: a universal 3D molecular representation learning framework. In *Proc. 11th Int. Conf. Learning Representations (ICLR)* (2023). OpenReview: 6K2RM6wVqKu.
- [19] Seidl, P., Vall, A., Hochreiter, S. & Klambauer, G. Enhancing activity prediction models in drug discovery with the ability to understand human language (CLAMP). In *Proc. 40th Int. Conf. Machine Learning*, PMLR **202**, 30458–30490 (2023).
- [20] Chen, T. & Guestrin, C. XGBoost: a scalable tree boosting system. In *Proc. 22nd ACM SIGKDD Conf. Knowledge Discovery and Data Mining*, 785–794 (2016). doi:10.1145/2939672.2939785
- [21] Bemis, G. W. & Murcko, M. A. The properties of known drugs. 1. Molecular frameworks. *J. Med. Chem.* **39**, 2887–2893 (1996). doi:10.1021/jm9602928
- [22] Wu, Z. *et al.* MoleculeNet: a benchmark for molecular machine learning. *Chem. Sci.* **9**, 513–530 (2018). doi:10.1039/C7SC02664A
- [23] Wallach, I. & Heifets, A. Most ligand-based classification benchmarks reward memorization rather than generalization. *J. Chem. Inf. Model.* **58**, 916–932 (2018). doi:10.1021/acs.jcim.7b00403

- 286 [24] Saito, T. & Rehmsmeier, M. The precision-recall plot is more informative than the ROC plot
287 when evaluating binary classifiers on imbalanced datasets. *PLoS ONE* **10**, e0118432 (2015).
288 doi:10.1371/journal.pone.0118432
- 289 [25] Truchon, J.-F. & Bayly, C. I. Evaluating virtual screening methods: good and bad metrics for the “early
290 recognition” problem. *J. Chem. Inf. Model.* **47**, 488–508 (2007). doi:10.1021/ci600426e
- 291 [26] Yang, Y., Pan, Z., Sun, J., Welch, J. & Klionsky, D. J. Autophagy and machine learn-
292 ing: unanswered questions. *Biochim. Biophys. Acta Mol. Basis Dis.* **1870**, 167263 (2024).
293 doi:10.1016/j.bbadis.2024.167263
- 294 [27] Huang, F., Guo, W., Chen, L., Feng, K., Huang, T. & Cai, Y.-D. Identifying autophagy-associated
295 proteins and chemicals with a random walk-based method within heterogeneous interaction network.
296 *Front. Biosci. (Landmark Ed.)* **29**, 21 (2024). doi:10.31083/j.fbl2901021
- 297 [28] Landrum, G. RDKit: open-source cheminformatics (software). <https://www.rdkit.org>